

Referee Report on “Leverage Risk and Investment: The Case of Gold Clauses in the 1930s”

Review of Financial Studies, MS # 20230176

This paper explores the effect on U.S. firm investment in the 1930s of the abrogation of “gold clauses” in 1933, which gave investors the free option to decide whether corporate debts were paid in gold or U.S. dollars (USD). The movement in the official price of gold for the first time in 100 years meant that if the abrogation were overturned the firms could have been exposed to a 69% increase in bond payments. The authors argue the analysis is intended to contribute to two important modern literatures: (1) the effect of “leverage risk”, such as those caused by exchange rate fluctuations and deflation; (2) investment under uncertainty.

They use hand-collected firm-level annual financial statements from the Moody’s Industrial Manuals from 1930-1936 for all firms with publicly traded common stock on the New York Stock Exchange. Since virtually all corporate bonds at this time had gold clauses and few other firm liabilities would, and firms that issue bonds are typically larger, it is not surprising that the authors find that firms with gold clauses in their debt were much larger and more highly levered. Over the two-year period of 1931 and 1932 though they don’t find any average difference in net % investment for those firms with debt gold clauses. By contrast, in the two-year period of 1933 and 1934, when there may have been additional “leverage risk”, they find that these firms had lower net investment and higher equity payouts. By the two-year period of 1935 and 1936, after the supreme court upheld the elimination of gold clauses, there is no longer a difference in net investment between these firms.

Since a natural concern is differential responses of these firms for reasons other than gold clauses, the authors show that their findings hold controlling for a broad set of co-variates including firm size, Q, leverage, liquidity, and profitability, as well as industry-year fixed effects. Findings are robust to the exact definitions of the variables of interest and are not driven by changes in profit, the proportion of liabilities that are bank debt, or firms with bonds maturing over the period of interest. Preferred stock was quite common at this time and are similar to a long-run liability, but didn’t include gold clauses. The authors use these, plus any bonds without gold clauses, as a placebo test and don’t find any effect on firm investment in the 1933 and 1934 periods. Taking their estimated effects to the data they estimate that, putting aside general equilibrium considerations, a substantial proportion of the divestment in the early 1930s among firms with gold clauses came from the leverage risk implicit in those clauses.

Taken together the authors argue that they provide evidence on the effects leverage risk can have on corporate investment with important implications for policies which could cause capital structure uncertainty.

Major Comments

1. The authors conduct a placebo test which includes preferred stock and “bond amount outstanding without gold clauses in total liabilities”. My impression though is that virtually all corporate bonds of any importance/size at this time included gold clauses. What percent of corporate bonds (by # and \$) in this sample are 100% confirmed to not have gold clauses? I would think that number, especially in \$ terms, is tiny. On a related note, what percent of all gold clause liabilities, as measured in this paper, come from corporate bonds vs any other liabilities (# and \$ terms)? Again, my impression

from looking through the Moody's Manuals is that virtually all the information on inclusion of gold clauses would be from long-term corporate bonds. If I'm mistaken and there is a lot of other variation it would be helpful for the authors to clarify that and demonstrate it in the text. It would also suggest that using the % of long-term corporate bonds with gold clauses, would be a much more natural workhorse empirical design and I would encourage the authors to use that. If, however, as appears to be the case - nearly all the variation in gold clause liabilities is variation in the amount of bonds - that needs to be stated and shown explicitly to the reader. To understand the validity of the design it is important to know that the primary source of variation is comparing firms with and without bonds, or with differing amount of bonds relative to their other liabilities, as opposed to firms who happened to include gold clauses or not in their bonds.

2. My biggest concern is the plausibility of the identifying assumptions within the research design of this paper. The implicit assumption here for these estimated effects to be valid is that over the two-year period of 1933 and 1934, the only difference of any importance driving variation in the change in net investment for firms with and without long-term corporate bonds (or more vs. less bonded debt as a % of total liabilities) is leverage risk from the gold clauses they have. This is despite the fact that firms with gold clauses - aka those with bonded debt - are on average more than twice as large, more than 50% more highly levered, have higher payouts, have more non-current assets, and lower profits. While it is comforting that the authors show that their findings are largely unchanged controlling both linearly and non-parametrically for many of these observed differences, it seems almost certain these firms differ on dimensions not analyzed/observed by the authors as well. Since the period examined is the aftermath of the Great Depression and includes the "first 100 days" period of FDR presidency, the establishment of the SEC, the Glass-Steagall Act, and huge numbers of other regulatory changes, market fluctuations, and re-organizations it is likely one of the more challenging periods in economic history to argue for the absence of concurrent confounds. With that in mind, I think there are a number of dimensions along which the authors could try to improve how they address these concerns:
 - a. Controls for observables: Since these firms differ in terms of their 1931-1932 payout/fixed capital it would be helpful to control for that interacted with the year linearly and non-parametrically just as is done for the other pretreatment values in Table 6. The following additional variables may not need to be in the main paper, but at a minimum in the appendix it would be helpful to include stock return volatility, Fama-French factor beta loadings, sales/assets, and location controls (ex. NYC vs. not). It would also be useful to see the full distribution of 1931-1932 variables for $d > 0$, $d = 0$, $d < 0.44$, and $d > 0.44$, even if it is just in an appendix.
 - b. Differences-in-Differences (DiD) Design: One of the primary arguments the authors put forth for why selection on unobservables is unlikely to be an issue is the evidence from their DiD design. That being said, the current implementation doesn't appear particularly compelling and I would encourage the authors to improve this substantially. It would be helpful to see the actual DiD coefficients plotted annually at a minimum all the years the authors state they already have data for (1930-1936). The way things are presented now, in just two-year aggregated increments, it is almost impossible to assess the validity of the DiD assumptions. Even more useful would be for them to extend their dataset to cover a few additional years both before and after the 1933 abrogation and 1935 supreme court decision to show that

- firms with gold clauses responded with net investment similarly in all years, except the two "treated" years of interest in 1933 and 1934. There is obviously no hard rule here, but something like 1925-1940 would at least be somewhat more convincing if DiD coefficients are plotted year-by-year and it is only 1933 and 1934 which are each statistically significantly lower, while all other years there is no differential net investment, relative to the omitted period.
- c. Secondary market data: Rather than looking at firms with publicly listed common stock on the NYSE (aka what is in CRSP) as a convenience sample, one justification is that there are a number of approaches using secondary market data that could help validate the research design. Monthly estimates of equity volatilities, average returns, volume of trading, bid-ask spreads, Fama-French factor loadings, et al. could all provide evidence of no differential trends in gold clause firms in the months prior to the events of interest, and a reversal that lines of up with what would be expected. For some of this period Edwards et al. (2015) even has measures of market implied probability of the gold clause "mattering" based on the spread between treasuries with and without the clause. Higher frequency data, like would be available with secondary market equity prices, could help validate that effects line up with changes in the more cleanly identified movements in Edwards et al. (2015). Dividend payouts are also higher frequency than annual and not paid out at the same time of year for all firms. That should provide additional higher frequency evidence on whether dividend payouts really change at the times that are expected given the events of interest.
 - d. % of long-term debt with gold clauses: I don't think there are, but if there are enough firms with bonds without gold clauses then it seems reasonable that the main specification should probably be the % of bonds (among those firms with bonds) outstanding with gold clauses. Given that the above is unlikely the most convincing evidence is likely the absence of any effects for preferred stock. Given that it would be helpful to see the results using just variation in the % of long-term debt (preferred stock + bonds) with gold clauses among those with either preferred stock or bonds. Just like with the primary design it would be helpful to see how within the sub-group those with or without any gold clauses (and pre period correlations with covariates) differ along other dimensions. Then historical discussions of why preferred don't include gold clauses and what drove decisions to issue one vs the other (ex. should we be concerned about differential exposure to the interest tax shield?). Any concerns addressed here are relevant for the original primary design, but alleviates many of the other concerns. In fact, I would argue even ex-ante this source of variation is much more likely to have plausible causal interpretation and the authors should seriously consider if an approach like this would be a more appropriate primary specification.
3. Another substantial concern is the lack of clarity on what exactly "leverage risk" is, is it new, what it would predict, and if those differentially predictions are really demonstrated empirically. The authors mention debt overhang and also models of uncertainty, but those seem like they might generate completely opposing predictions. If there are costs of financial distress then it seems likely that firms would want to build-up cash, rather than engage in equity payouts, if they are concerned about future uncertainty (aka an increased probability of needing to make larger debt payments to avoid distress). On the other hand, if it is about debt overhang then perhaps as the authors indicate

firms could want to increase equity payouts. If debt overhang is the mechanism though, then there does already exist some empirical evidence in that space (ex. Bennett et al. 2018 and Wittry 2018 as recent examples) and it isn't clear in the way things are written now that leverage risk is empirically distinguishable from debt overhang in this setting. As the authors note in this setting conditional on the abrogation being overturned there could be a 69% increase in bond payments for exposed firms just after perhaps the most severe economic downturn in U.S. history. It seems entirely plausible this has a substantial effect on the probability of bankruptcy, and therefore debt overhang. Firms always have some "leverage risk" since the future value of the asset side of the balance sheet is always uncertain and that could generate debt overhang if firms are sufficiently levered. What then are the differential predictions when there is "leverage risk" coming from uncertainty in the future liability and how do the authors show evidence of that? If there aren't differential predictions, then it might be more appropriate to frame this paper as additional evidence on the effects of debt overhang on corporate investment and then compare/contrast much more with the existing literature/evidence in that space.

Minor Comments

1. It would be helpful if the authors explored the equity payout results in a bit more detail. I would encourage the authors to try a number of different definitions including the % change in payouts, dividend yield, % of profits paid out as dividends, and break out changes coming from cash dividends vs. share repurchases and intensive vs. extensive margins (not paying out at all vs. paying out only somewhat).
2. It would be useful to see the coefficient on d in Table 4 column 1 of Panels A and B.
3. What are the number of clusters in these regressions?
4. What do the results look like using just the $d > 0$ indicator variable that matches more closely to what is done in Figure 1?
5. What are the correlations (and statistical significance) between d and co-variates in 1931-1932 period?
6. How do the authors reconcile Figure 1 and results from the rest of the paper? Figure 1 shows as the authors note that "[w]e find that leverage risk led public firms to delay investment in fixed capital for two years." That would suggest that not only does the period in 1933-1934 have a lower net investment rate than 1931-1932, but that 1935-1936 has a higher net investment rate than 1931-1932 and enough higher that the cumulative effects are 0 over this whole period. In other words, no investment is lost, just delayed. By contrast, Table 2 and 4 both suggest that 1935-1936 net investment is going to be basically identical to that in 1931-1932. That would suggest there is a cumulative effect as 1935-1936 net investment does not offset the decline in 1933-1934. It would be helpful for the authors to explicitly test 1935-1936 relative to 1931-1932 and provide some discussion for reader about which findings (the table or figure) seem most relevant and what the delay vs loss means in terms of the theories being tested (ex. Mello and Parsons 1992 would seem

to suggest a delay). One natural prediction would seem to be that the reduction in uncertainty in 1935-1936 should lead to the same net investment difference as would occur in 1931-1932, plus some additional positive net investment for those treated firms. The current evidence in the main paper/tables doesn't seem to support that conclusion.

7. If the gold clauses hurt firm value, how come we don't see that in Tobin's Q? It would be helpful to show explicitly that the equity value of these firms are hurt by the risk during this period or if the authors find it isn't provide a plausible explanation for why.
8. How should we reconcile the findings in this paper with those in Figure 4 of Edwards et al. (2015)? They find no effect of gold clauses until 1934, while this paper finds evidence in 1933. They find no reduction in the risks of the gold clause in 1935, while this paper suggests there are. It would be good to include a detailed discussion of these divergent assumptions/findings and why the authors don't believe they are problematic. At its surface high frequency data on prices for otherwise virtually identical bonds with and without gold clauses trading on the same days seems like a fairly clean approach to examine the pricing effects of gold clauses, so it seems important to reconcile with those findings.

References

- Bennett, Benjamin, Todd Milbourn, and Zexi Wang (2018). Corporate Investment Under the Cloud of Litigation. Working Paper.
- Mello, Antonio S., and John E. Parsons (1992), Measuring the agency cost of debt, *Journal of Finance* 47, 1887–1904.
- Wittry, Michael (2018). (Debt) Overhang: Evidence from Resource Extraction. Working paper.